



1. Description

1.1. Project

Project Name	MusicPlayerFirmware
Board Name	NUCLEO-G031K8
Generated with:	STM32CubeMX 6.16.1
Date	01/07/2026

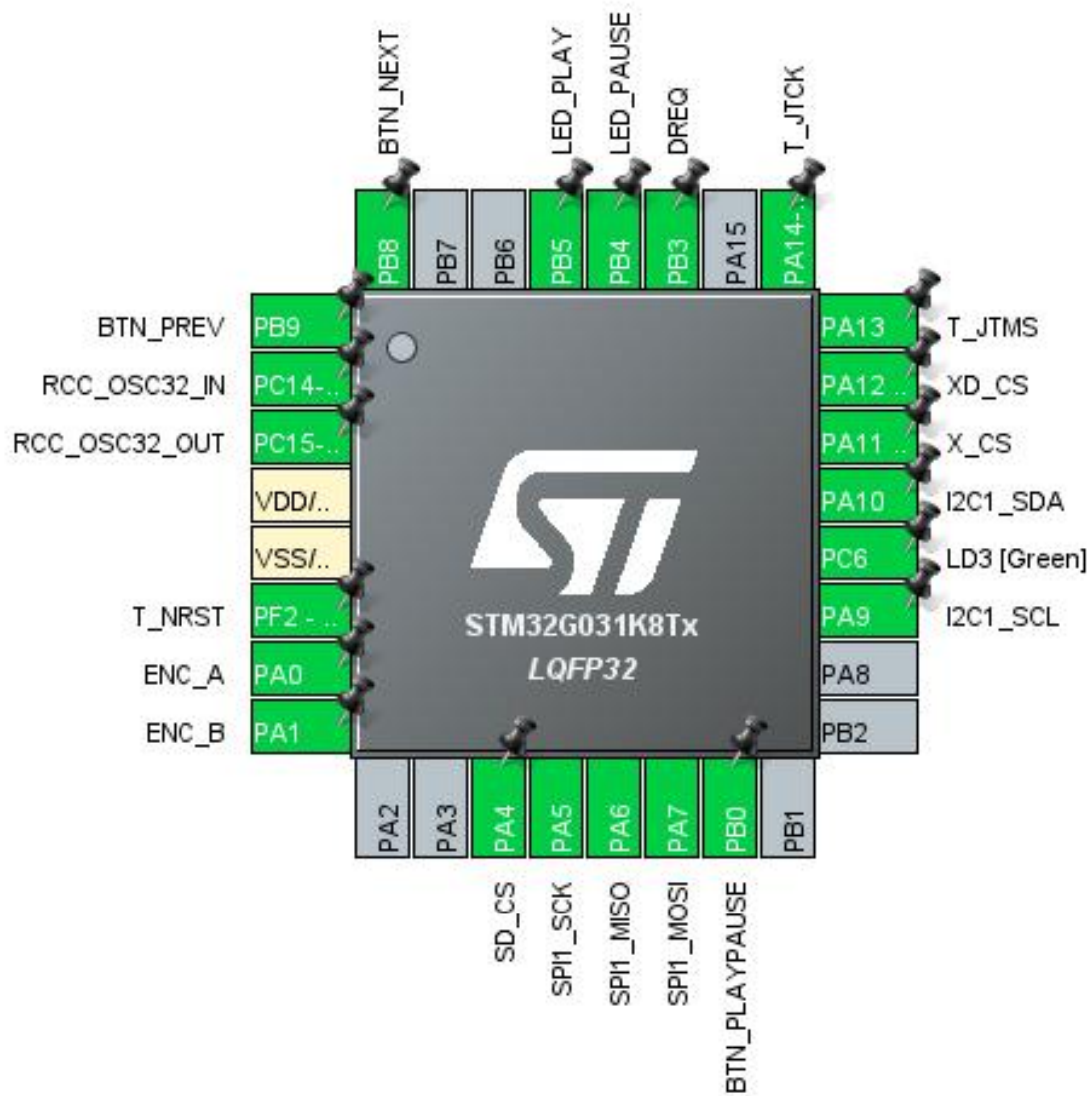
1.2. MCU

MCU Series	STM32G0
MCU Line	STM32G0x1
MCU name	STM32G031K8Tx
MCU Package	LQFP32
MCU Pin number	32

1.3. Core(s) information

Core(s)	ARM Cortex-M0+
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP32	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PB9	I/O	GPIO_EXTI9	BTN_PREV
2	PC14-OSC32_IN (PC14)	I/O	RCC_OSC32_IN	
3	PC15-OSC32_OUT (PC15)	I/O	RCC_OSC32_OUT	
4	VDD/VDDA	Power		
5	VSS/VSSA	Power		
6	PF2 - NRST	I/O	GPIO_EXTI2	T_NRST
7	PA0	I/O	TIM2_CH1	ENC_A
8	PA1	I/O	TIM2_CH2	ENC_B
11	PA4 *	I/O	GPIO_Output	SD_CS
12	PA5	I/O	SPI1_SCK	SPI1_SCK
13	PA6	I/O	SPI1_MISO	SPI1_MISO
14	PA7	I/O	SPI1_MOSI	SPI1_MOSI
15	PB0	I/O	GPIO_EXTI0	BTN_PLAYPAUSE
19	PA9	I/O	I2C1_SCL	
20	PC6 *	I/O	GPIO_Output	LD3 [Green]
21	PA10	I/O	I2C1_SDA	
22	PA11 [PA9] *	I/O	GPIO_Output	X_CS
23	PA12 [PA10] *	I/O	GPIO_Output	XD_CS
24	PA13	I/O	SYS_SWDIO	T_JTMS
25	PA14-BOOT0	I/O	SYS_SWCLK	T_JTCK
27	PB3 *	I/O	GPIO_Input	DREQ
28	PB4	I/O	TIM3_CH1	LED_PAUSE
29	PB5	I/O	TIM3_CH2	LED_PLAY
32	PB8	I/O	GPIO_EXTI8	BTN_NEXT

* The pin is affected with an I/O function

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32G0
Line	STM32G0x1
MCU	STM32G031K8Tx
Datasheet	DS12992 Rev0

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(AAA700)
Capacity	700.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	10.0 mA
Max Pulse Current	30.0 mA
Cells in series	1
Cells in parallel	1

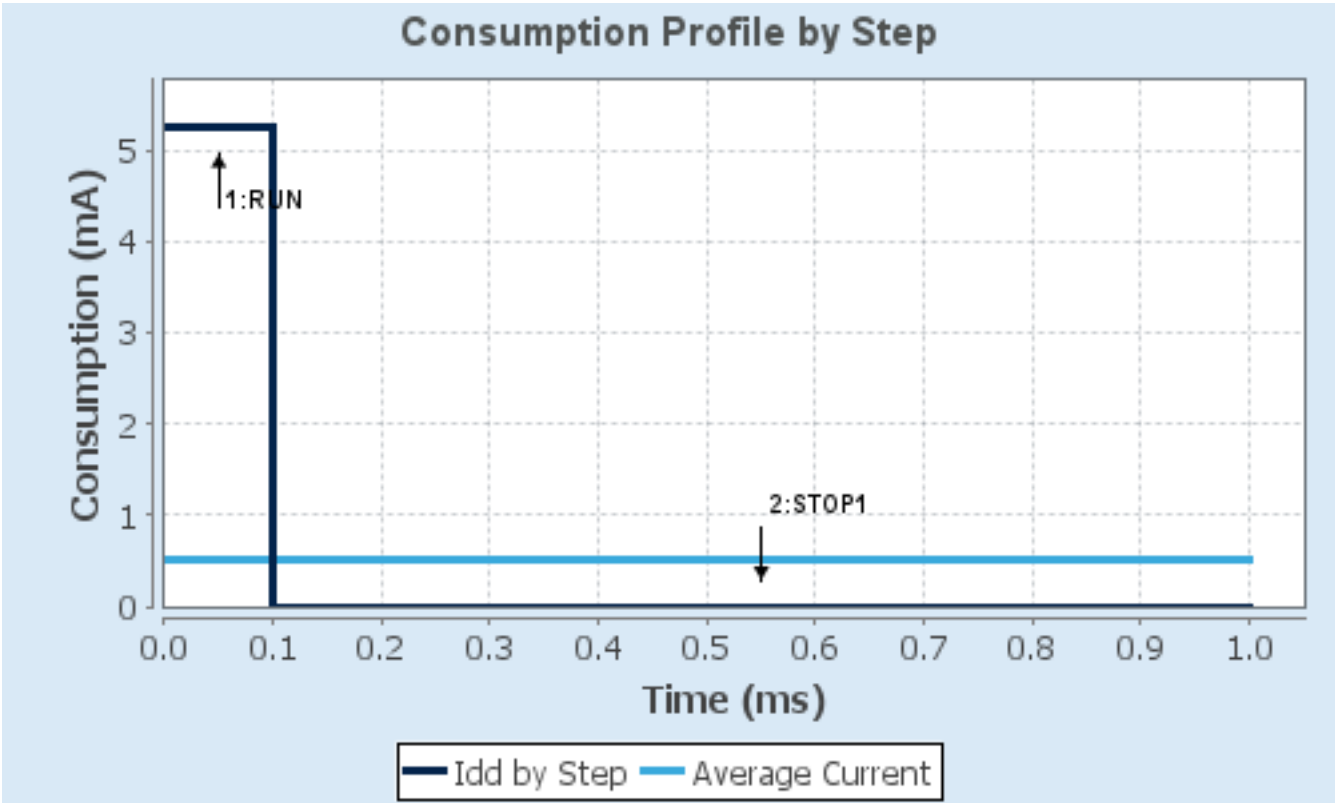
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP1
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-High	Range1-High
Fetch Type	FLASH	Flash-PowerDown
CPU Frequency	64 MHz	16 MHz
Clock Configuration	HSI PLL	HSI
Clock Source Frequency	16 MHz	16 MHz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	5.25 mA	3.36 μ A
Duration	0.1 ms	0.9 ms
DMIPS	80.0	0.0
Ta Max	128.8	130
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	528.02 μ A
Battery Life	1 month, 24 days, 17 hours	Average DMIPS	80.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	MusicPlayerFirmware
Project Folder	C:\Users\josh\PersonalProjects\EmbeddedProjects\MusicPlayer\MusicPlayerFir
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_G0 V1.6.2
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_I2C1_Init	I2C1
5	MX_SPI1_Init	SPI1
6	MX_TIM3_Init	TIM3
7	MX_TIM2_Init	TIM2
8	MX_FATFS_Init	FATFS

3. Peripherals and Middlewares Configuration

3.1. I2C1

I2C: I2C

3.1.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Fast Mode *
I2C Speed Frequency (KHz)	400
Rise Time (ns)	100
Fall Time (ns)	100
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x00300617 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	60 *

3.2. RCC

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.2.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	2 WS (3 CPU cycle)

RCC Parameters:

HSI Calibration Value	64
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
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Peripherals Clock Configuration:

Generate the peripherals clock configuration TRUE

3.3. SPI1

Mode: Full-Duplex Master

3.3.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	8 *
Baud Rate	8.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software

3.4. SYS

mode: Debug

Timebase Source: SysTick

3.5. TIM2

Combined Channels: Encoder Mode

3.5.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode

____ Parameters for Channel 1 ____

Polarity

IC Selection

Prescaler Division Ratio

Input Filter

____ Parameters for Channel 2 ____

Polarity

IC Selection

Prescaler Division Ratio

Input Filter

Encoder Mode TI1 and TI2 *

Falling Edge *

Direct

No division

0

Falling Edge *

Direct

No division

0

3.6. TIM3

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

3.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	63 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	999 *
Internal Clock Division (CKD)	No Division
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable
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PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	250 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	250 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.7. FATFS

mode: User-defined

3.7.1. Set Defines:

Version:

FATFS version	R0.12c
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Function Parameters:

FS_READONLY (Read-only mode)	Disabled
FS_MINIMIZE (Minimization level)	Enabled with 6 functions removed *
USE_STRFUNC (String functions)	Disabled *
USE_FIND (Find functions)	Disabled
USE_MKFS (Make filesystem function)	Disabled *
USE_FASTSEEK (Fast seek function)	Disabled *
USE_EXPAND (Use f_expand function)	Disabled
USE_CHMOD (Change attributes function)	Disabled
USE_LABEL (Volume label functions)	Disabled
USE_FORWARD (Forward function)	Disabled

Locale and Namespace Parameters:

CODE_PAGE (Code page on target)	Latin 1
USE_LFN (Use Long Filename)	Disabled
MAX_LFN (Max Long Filename)	255
LFN_UNICODE (Enable Unicode)	ANSI/OEM
STRF_ENCODE (Character encoding)	ANSI/OEM *
FS_RPATH (Relative Path)	Disabled

Physical Drive Parameters:

VOLUMES (Logical drives)	1
MAX_SS (Maximum Sector Size)	512
MIN_SS (Minimum Sector Size)	512
MULTI_PARTITION (Volume partitions feature)	Disabled
USE_TRIM (Erase feature)	Disabled
FS_NOFSINFO (Force full FAT scan)	0

System Parameters:

FS_TINY (Tiny mode)	
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	Enabled *
FS_EXFAT (Support of exFAT file system)	Disabled
FS_NORTC (Timestamp feature)	Fixed timestamp *
NORTC_YEAR (Year for timestamp)	2026 *
NORTC_MON (Month for timestamp)	1 *
NORTC_MDAY (Day for timestamp)	1 *
FS_REENTRANT (Re-Entrancy)	Disabled
FS_TIMEOUT (Timeout ticks)	1000
FS_LOCK (Number of files opened simultaneously)	1 *

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
I2C1	PA9	I2C1_SCL	Alternate Function Open Drain	Pull-up *	Low	
	PA10	I2C1_SDA	Alternate Function Open Drain	Pull-up *	Low	
RCC	PC14-OSC32_IN (PC14)	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT (PC15)	RCC_OSC32_OUT	n/a	n/a	n/a	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	SPI1_SCK
	PA6	SPI1_MISO	Alternate Function Push Pull	Pull-up *	Low	SPI1_MISO
	PA7	SPI1_MOSI	Alternate Function Push Pull	Pull-up *	Low	SPI1_MOSI
SYS	PA13	SYS_SWDI0	n/a	n/a	n/a	T_JTMS
	PA14-BOOT0	SYS_SWCLK	n/a	n/a	n/a	T_JTCK
TIM2	PA0	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC_A
	PA1	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC_B
TIM3	PB4	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_PAUSE
	PB5	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	LED_PLAY
GPIO	PB9	GPIO_EXTI9	External Interrupt Mode with Rising edge trigger detection	Pull-down *	n/a	BTN_PREV
	PF2 - NRST	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	T_NRST
	PA4	GPIO_Output	Output Push Pull	Pull-up *	Low	SD_CS
	PB0	GPIO_EXTI0	External Interrupt Mode with Rising edge trigger detection	Pull-down *	n/a	BTN_PLAYPAUSE
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD3 [Green]
	PA11 [PA9]	GPIO_Output	Output Push Pull	Pull-up *	Low	X_CS
	PA12 [PA10]	GPIO_Output	Output Push Pull	Pull-up *	Low	XD_CS
	PB3	GPIO_Input	Input mode	Pull-down *	n/a	DREQ
	PB8	GPIO_EXTI8	External Interrupt Mode with Rising edge trigger detection	Pull-down *	n/a	BTN_NEXT

4.2. DMA configuration

DMA request	Stream	Direction	Priority
I2C1_TX	DMA1_Channel1	Memory To Peripheral	Medium *

I2C1_TX: DMA1_Channel1 DMA request Settings:

Mode: Normal
Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
System service call via SWI instruction	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	0	0
EXTI line 0 and line 1 interrupts	true	0	0
EXTI line 2 and line 3 interrupts	true	0	0
EXTI line 4 to 15 interrupts	true	0	0
DMA1 channel 1 interrupt	true	0	0
I2C1 event global interrupt / I2C1 wake-up interrupt through EXTI line 23	true	0	0
SPI1 global interrupt	true	0	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
System service call via SWI instruction	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
EXTI line 0 and line 1 interrupts	false	true	true
EXTI line 2 and line 3 interrupts	false	true	true
EXTI line 4 to 15 interrupts	false	true	true
DMA1 channel 1 interrupt	false	true	true
I2C1 event global interrupt / I2C1 wake-up interrupt through EXTI line 23	false	true	true
SPI1 global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Middleware					
FATFS ✓					
System Core	Analog	Timers	Connectivity	Multimedia	Computing
DMA ✓		TIM2 ✓	I2C1 ✓		
GPIO ✓		TIM3 ✓	SPI1 ✓		
NVIC ✓					
RCC ✓					
SYS ✓					

6. Docs & Resources

Type	Link
IBIS models	https://www.st.com/resource/en/ibis_model/stm32g0_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32g0-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
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Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
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for related Tools
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Application Notes https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpdp-sink-application-with-stm32cubemx-stmicroelectronics.pdf
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& Software

Application Notes https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
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Manuals	cortexm0-mcus-programming-manual-stmicroelectronics.pdf
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